1. Clinical Question & Core Hypothesis

Clinical Question

What molecular mechanisms determine whether immune structures translate into clinical benefit?

Working Hypothesis

In ERBB2-amplified breast cancers, TP53 mutation drives tumor-cell intrinsic transcriptional reprogramming that functionally diminishes anti-tumor immunity, such that high Tumor-Infiltrating Lymphocyte (TIL) density does not predict effective immune-mediated tumor control. Clinically, TP53-mutant/ERBB2-amplified tumors will show poorer therapy_response to HER2-targeted therapy compared with TP53-wild-type/ERBB2-amplified tumors, particularly in tumors with high TIL density.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	TP53, ERBB2
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

TP53: TP53 → mutated_in → Breast cancer (TP53 mutation as a core driver of genomic instability/transcriptional reprogramming) and (b) Breast cancer → predict → HER2 (ERBB2 amplification/activation as the therapeutic axis).

ERBB2: ERBB2/HER2 as a clinically actionable predictor and PD-L1/cancer-cell immune modulation as a mechanism that could specifically blunt HER2-targeted therapy efficacy (which relies in part on immune mechanisms such as ADCC), so a therapy-response proxy (HER2-targeted therapy response) is the most problem-relevant endpoint.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	CD274	Supporting modifier from knowledge graph
Secondary	Gene:genes	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: to the immune context and clinical outcome, without expanding the core hypothesis beyond the two anchor genes. Secondary modifiers (e.g., CD274/PD-L1 and broader immune-regulatory gene sets) will be analyzed to contextualize mechanism but are not required in the core hypothesis.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Gene expression does not equal protein activity; phospho-protein or functional assays are needed for confirmation.

Alternative Directions

- CD274: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- Gene:genes: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.